

**wellcome
connecting
science**

Pathway and network analysis

An introduction to pathway analysis with WikiPathways

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Maastricht Centre for Systems Biology and Bioinformatics (MaCSBio)

Computational Systems Biology for Digital Medicine, WCS course 2025

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Maastricht University



Interactive session

- Introduction pathways + WikiPathways
- Explanation and exploration of the dataset
 - Differential gene expression
 - Small cell lung cancer
- Pathway enrichment analysis
 - Pathway databases
 - Statistical analysis to identify affected pathways in dataset
- Pathway visualization
 - Data visualization and interpretation

Setup

- This will be an interactive session with switching between lecture / demo and practical steps.
- Read the explanations of the steps and discuss the output
→ try to understand and interpret the results
- **Discuss with each other**

Introduction



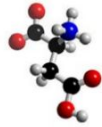
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Learning objectives

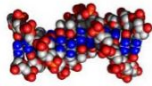
1. Introduction WikiPathways
2. Study the molecular mechanisms underlying Small Cell Lung Cancer
3. Biological insights into lung cancer from transcriptomics data
4. Tools for pathway analysis and visualization

Introduction

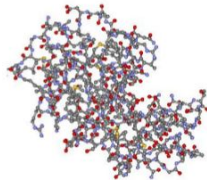
Molecules of life do not function in isolation



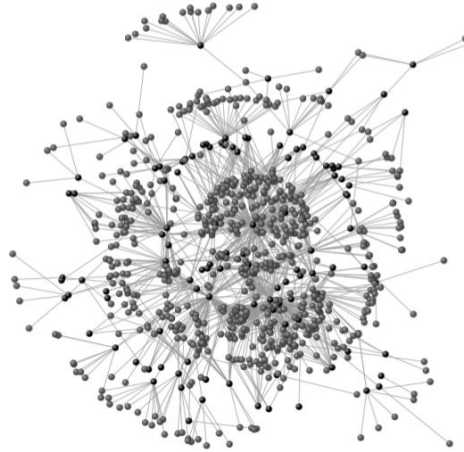
metabolites



genes

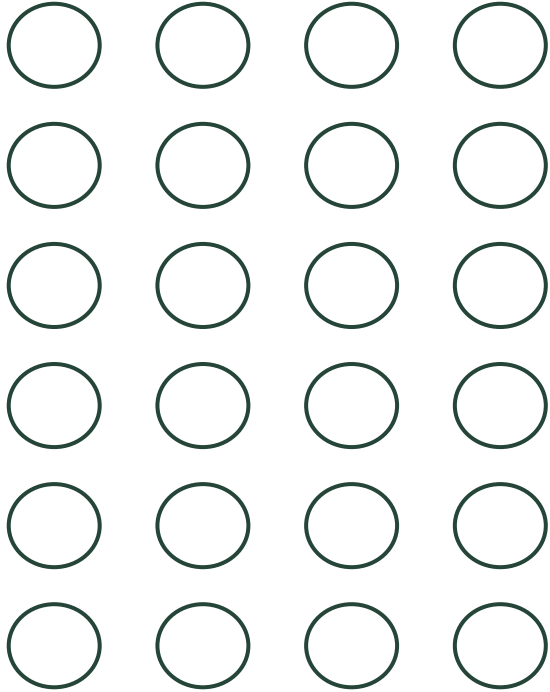


proteins



but form complex networks that define a cell

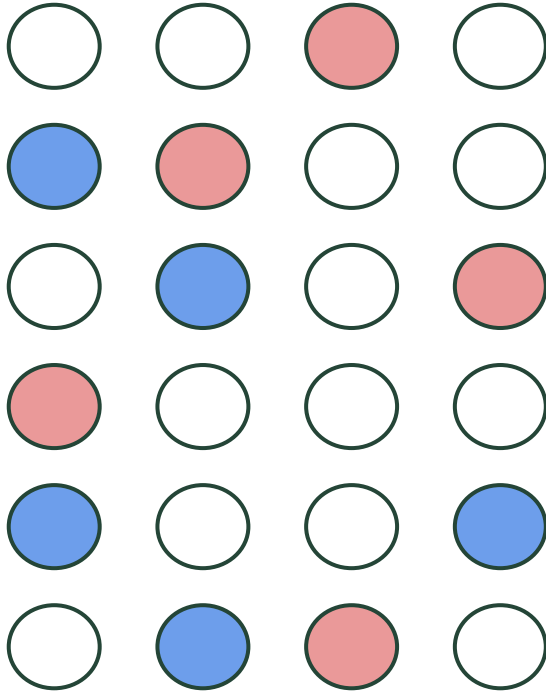
Introduction enrichment analysis



Quantify

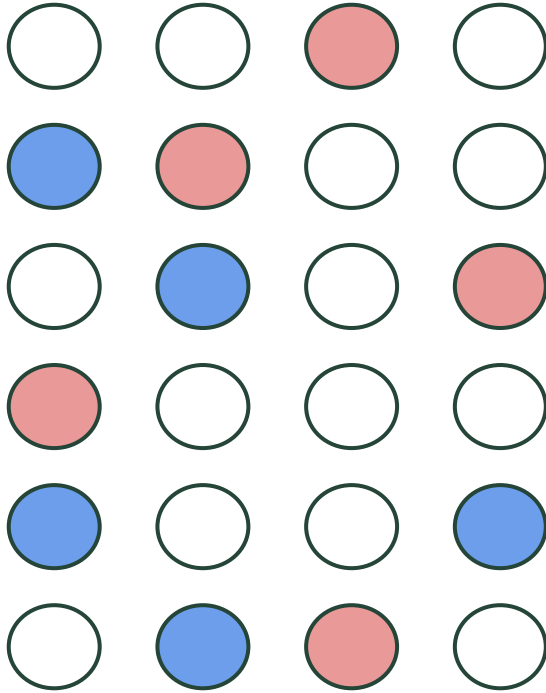
Isolated data points

Introduction enrichment analysis



Comparative statistics
Genes of interest
(up- or down-regulated genes)

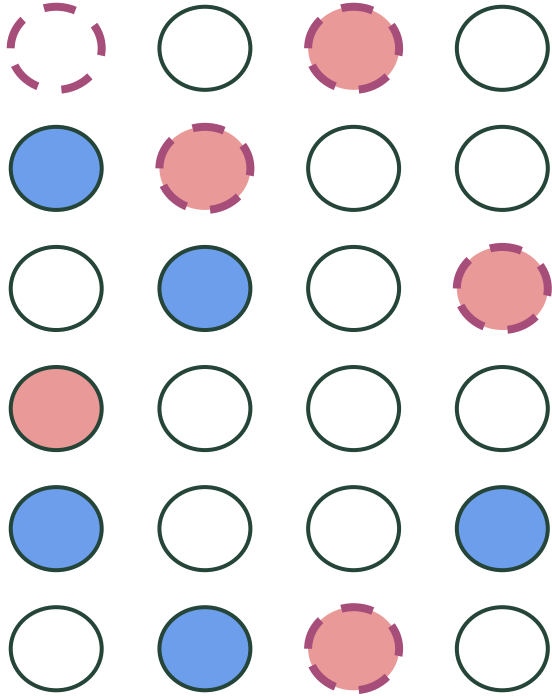
Introduction enrichment analysis



Enrichment analysis

Pre-defined gene sets →
functional groups

Introduction enrichment analysis

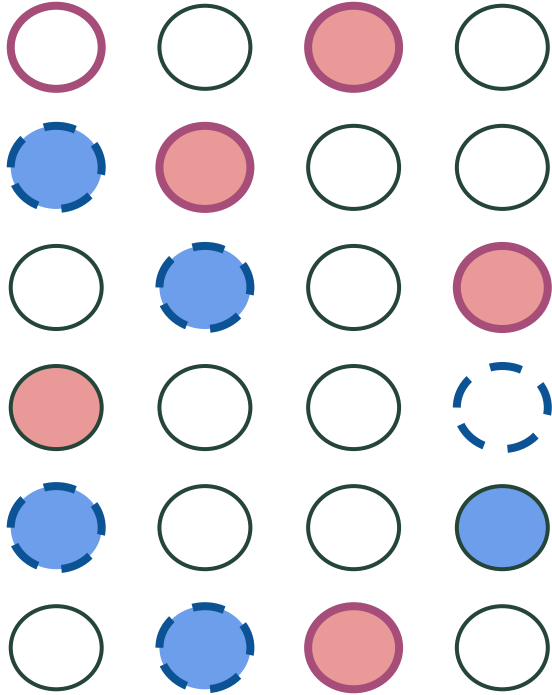


Enrichment analysis

Pre-defined gene sets →
functional groups



Introduction enrichment analysis



Enrichment analysis

Pre-defined gene sets → functional groups

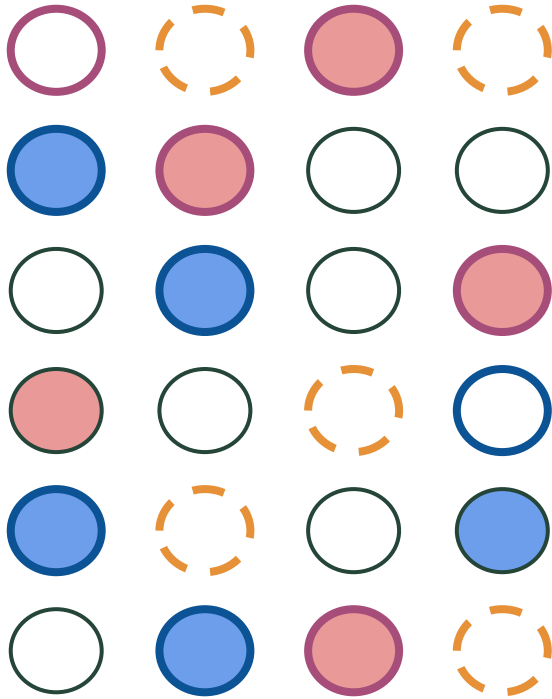


Apoptosis



Amino acid metabolism

Introduction enrichment analysis

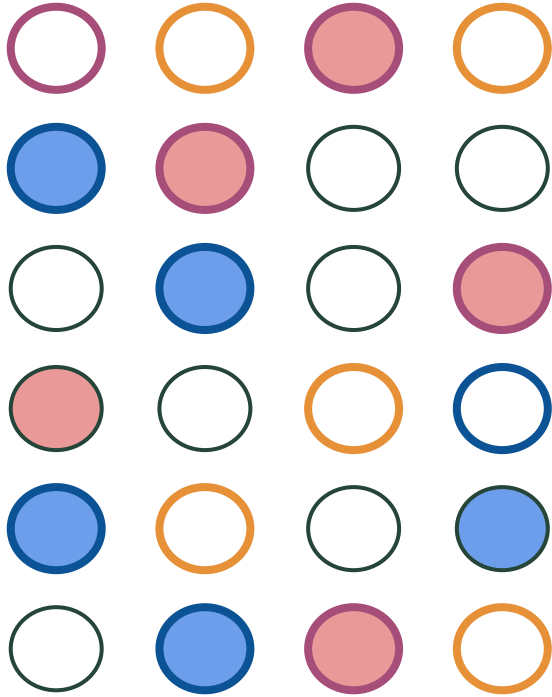


Enrichment analysis

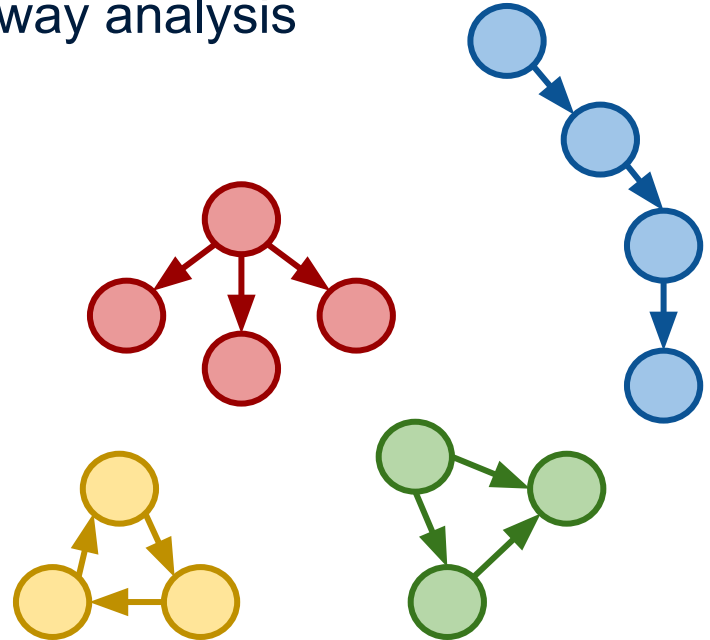
Pre-defined gene sets → functional groups



Introduction enrichment analysis



Enrichment analysis
Pathway analysis



Pathway = gene set with information about relationships

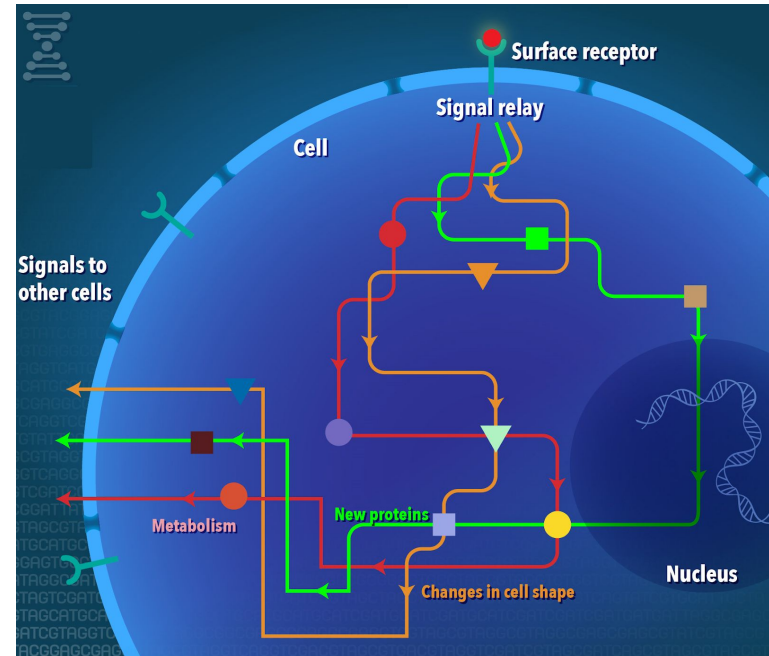
Introduction WikiPathways



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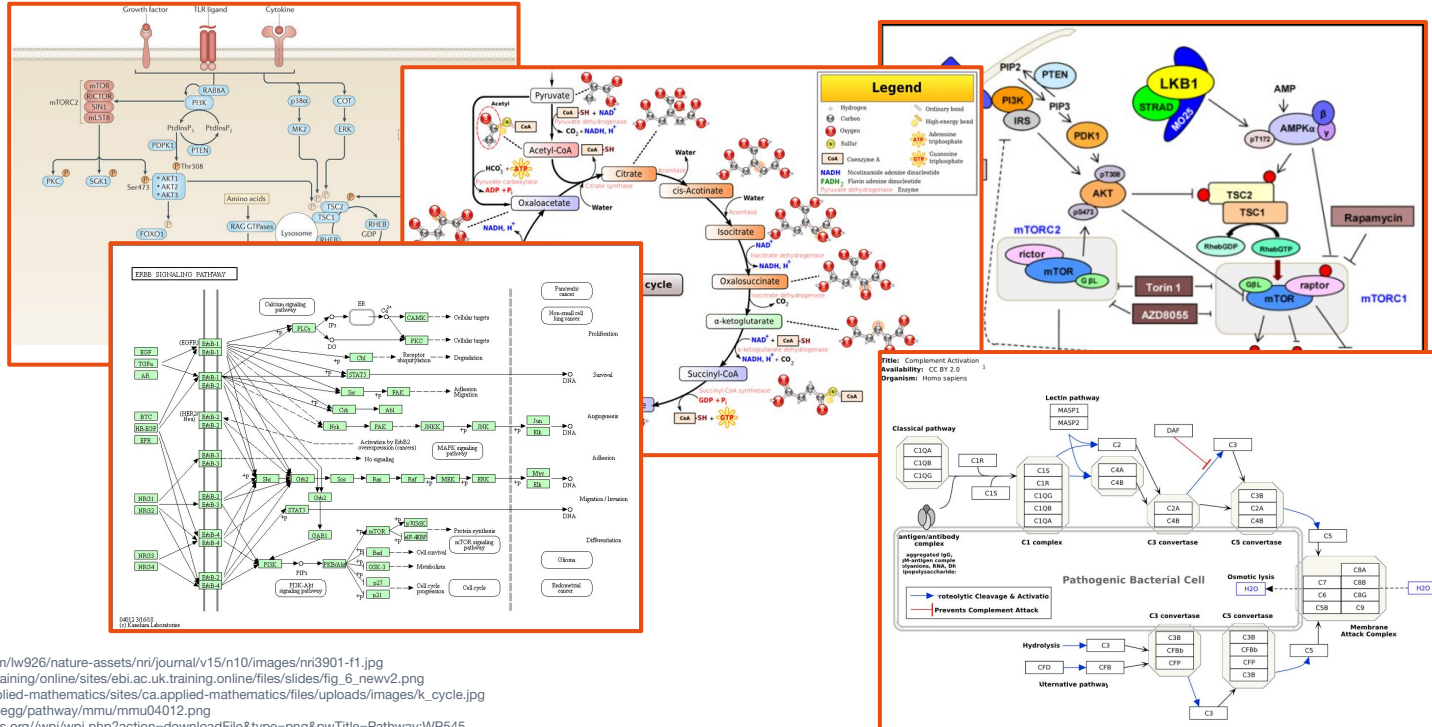
Biological pathways

- Signaling pathways
- Metabolic pathways
- Gene regulation pathways



Biological pathways

Pathway diagrams are found everywhere!



Biological pathways

Pathway diagrams are found everywhere!



Utility to biologists as conceptual models is obvious

Biological pathways

Pathway diagrams are found everywhere!

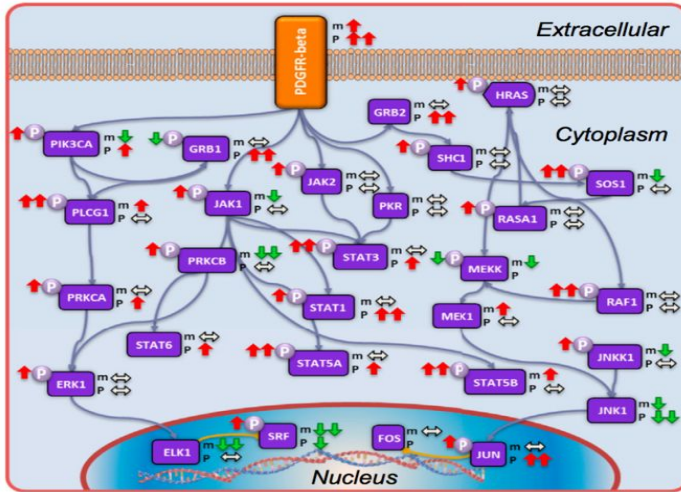


Utility to biologists as conceptual models is obvious

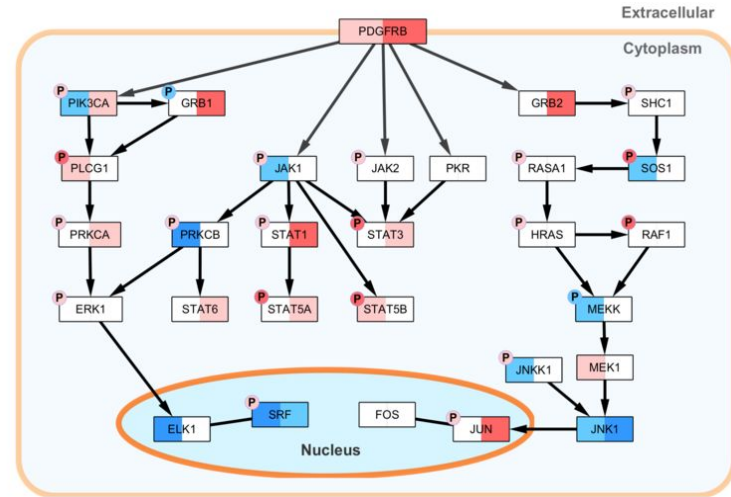


If modeled properly - immensely useful for computational analysis and interpretation of large-scale experimental data

Biological pathways



Static image
Zhang et al, Cell 2016



**PDGFR-beta pathway with
transcriptomic/phosphoproteomic data**
www.wikipathways.org/instance/WP3972

WikiPathways

- Experiment in community-based curation of biological pathways at Maastricht University together with the Gladstone Institutes in San Francisco



Too much data!

Difficult to keep knowledge up-to-date, accessible and integrated



Taking advantage of direct participation by a greater portion of the community
(crowdsourcing)

Pathway Databases

PDB Name	Pathway focus	URL	Y.O.R.	Standard formats
EcoCyc	M,S	biocyc.org	1995	SBML, BioPAX
KEGG	M,S,D	kegg.jp	1996	BioPAX
RegulonDB	GR	regulondb.ccg.unam.mx	1997	BioPAX
MetaCyc	M	metacyc.org	1999	SBML, BioPAX
STRINGDB	PPI	string-db.org	2000	PSI-MI
PANTHER	S,D,PS	pantherdb.org	2004	SBML, SBGN, BioPAX
Gene Ontology	PPI,M,S	geneontology.org	2000	
REACTOME	M,S,D	reactome.org	2005	SBML, SBGN, BioPAX, PSI-MI
MSigDb	M,S,GR	broadinstitute.org/gsea/msigdb	2005	
Ingenuity Knowledge Base*	PPI,PCI,M,S,GR,D	ingenuity.com	2005	
NCI PID	S,D	pid.nci.nih.gov	2006	BioPAX
WikiPathways	M,S,D	wikipathways.org	2008	BioPAX
Small Molecule Pathway DB	M,S	smpdb.ca	2009	SBML, BioPAX
ConsensusPathDB	PPI,PCI,M,S,GR	consensuspathdb.org	2009	BioPAX, PSI-MI
Pathway Commons	PPI,PCI,M,S	pathwaycommons.org	2010	BioPAX

*A brief example of the diversity of available PDBs found online. The second column shows the kind of biological focus pursued by each database: (PPI, protein-protein interactions; PCI, protein-compound interactions; M, metabolic; S, signaling; GR, gene regulation; D, diagrams; PS, protein sequence). The last column addresses the standard pathway languages adopted to provide data. Additionally, in the third column the links to web sites are supplied. YOR, Year of release. *Commercial database.*

WikiPathways

- Community-curated
- Collaborative
- Open

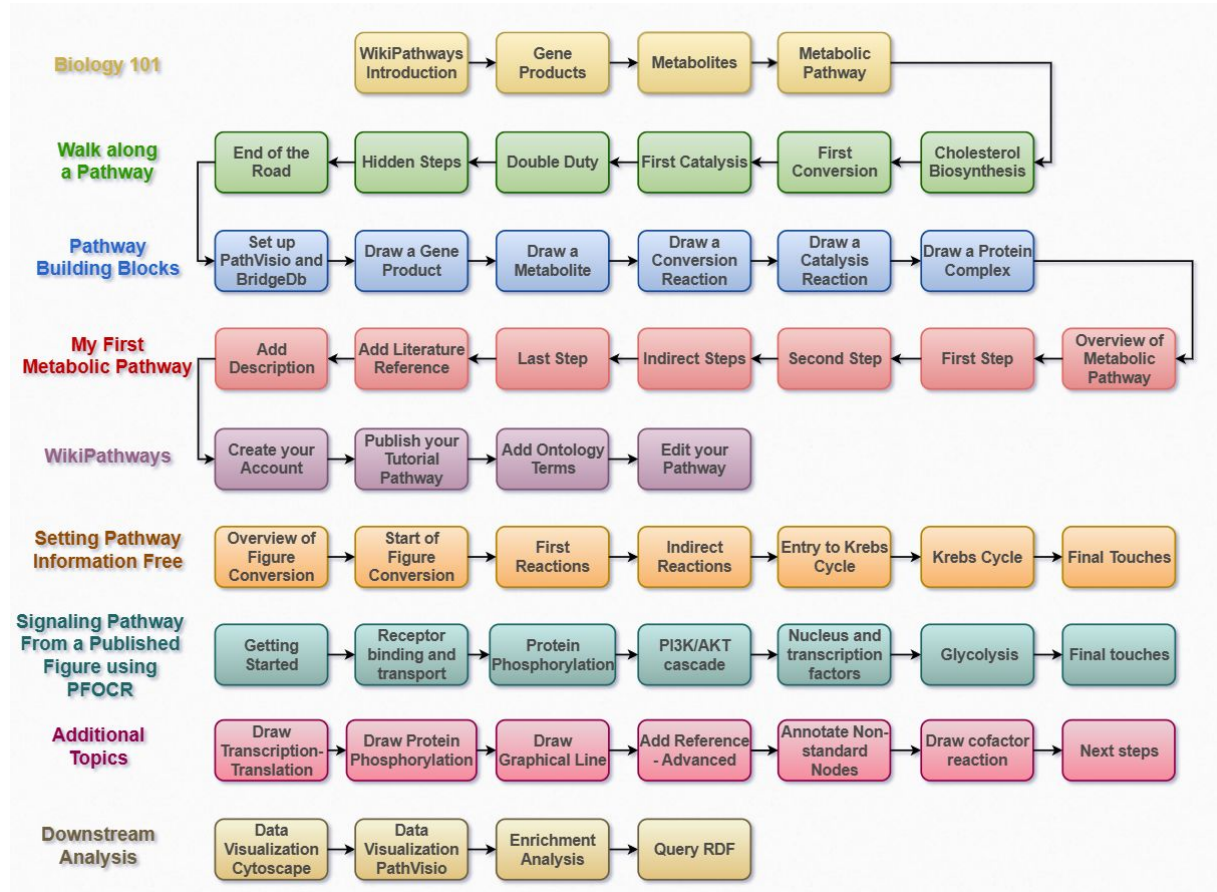
Content:

- ~2000 pathways
- 27 species
- 800+ editors

www.wikipathways.org

The screenshot displays the WikiPathways homepage. At the top, there is a search bar with the text "e.g., ACE or cancer" and an "Advanced search" link. Below the search bar is a "Go to Classic Site" button. The main navigation menu includes links for "About", "Search", "Browse", "Communities", "Download", "Analyze", "Cite", and "Help". A welcome message states: "Welcome to the new WikiPathways site! Please help report any issues you find." Below this, a description of WikiPathways is provided: "WikiPathways is an open science platform for biological pathways contributed, updated, and used by the research community." A "Read more" button is located below the description. The page is powered by "PathVisio" and "Bridges". A large, detailed biological pathway diagram is shown, featuring nodes such as "ACE2 (Q14875321)", "ACE", "KNG1", "Bradykinin", "BDKRB1", "BDKRB2", and "Prostacyclin". The diagram illustrates the signaling pathway involving ACE2 and ACE, leading to the release of Bradykinin and its subsequent effects on BDKRB1 and BDKRB2, which then lead to Prostacyclin production. A text box at the bottom of the diagram states: "Interact with diagrams with clickable genes, drugs, and pathways." Below the main content area, there is a section titled "Browse for Pathways" with the text: "Explore the full breadth and depth of pathway knowledge. Discover pathways of interest by organism, communities of domain experts, and ontology annotations." This section includes a grid of pathway thumbnails and a sidebar with filters for "Organisms", "Communities", and "Annotations".

WikiPathways academy



What can you do with pathway models?

1. Data visualization

- How is the pathway affected in my experiment?

2. Pathway statistics / enrichment analysis

- Which pathways are affected in my experiment?

3. Network analysis (optional extension)

- How is the pathway regulated by transcription factors / drugs?
- How is the pathway connected to other pathways (pathway crosstalk)?

4. Starting point for building your boolean/ODE models

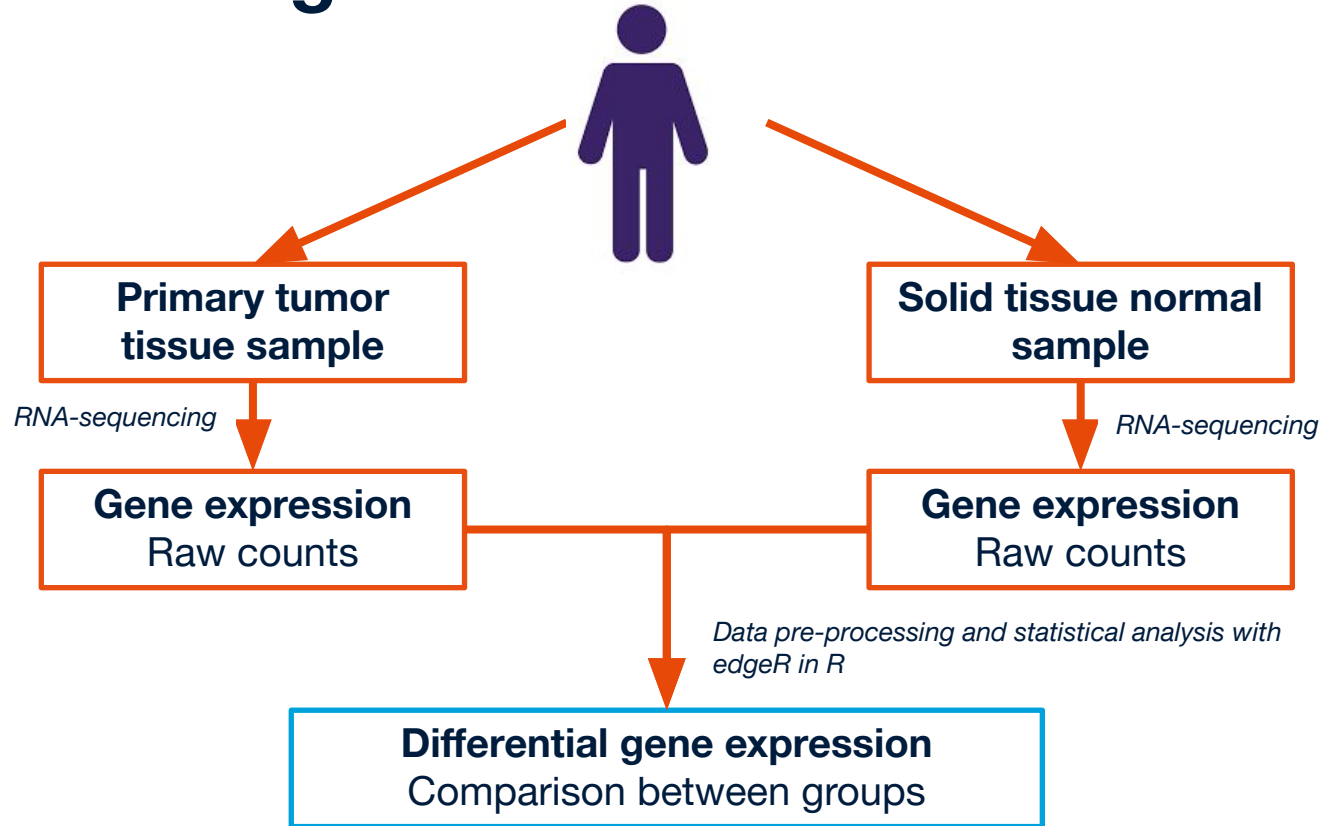
Explanation and exploration of the dataset



Dataset

- Jiang, *et al.* (2016) The RNAseq of 79 small cell lung cancer (sclc) and 7 normal control
- Publicly available dataset - GEO database
 - <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE60052>
- Comparing SCLC vs. healthy control

Dataset – lung cancer



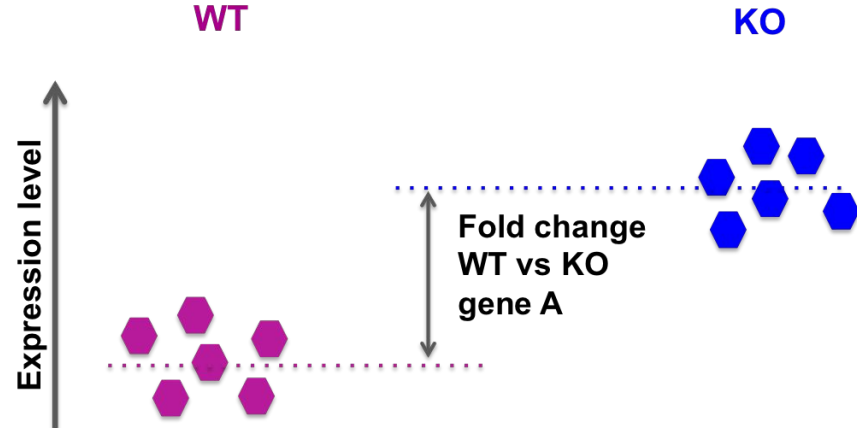
Dataset

GeneID	GeneName	log2FC	pvalue	padj
ENSG00000230657	PRB4	13.2739759	0.003920605	0.097861917

- GeneID □ identifier in online database
- GeneName □ official gene symbol
- log2FC □ log2 of fold change (ratio of the differences between cancer and healthy samples)
- pvalue □ significance level of comparison
- padj □ corrected P.Value for multiple testing

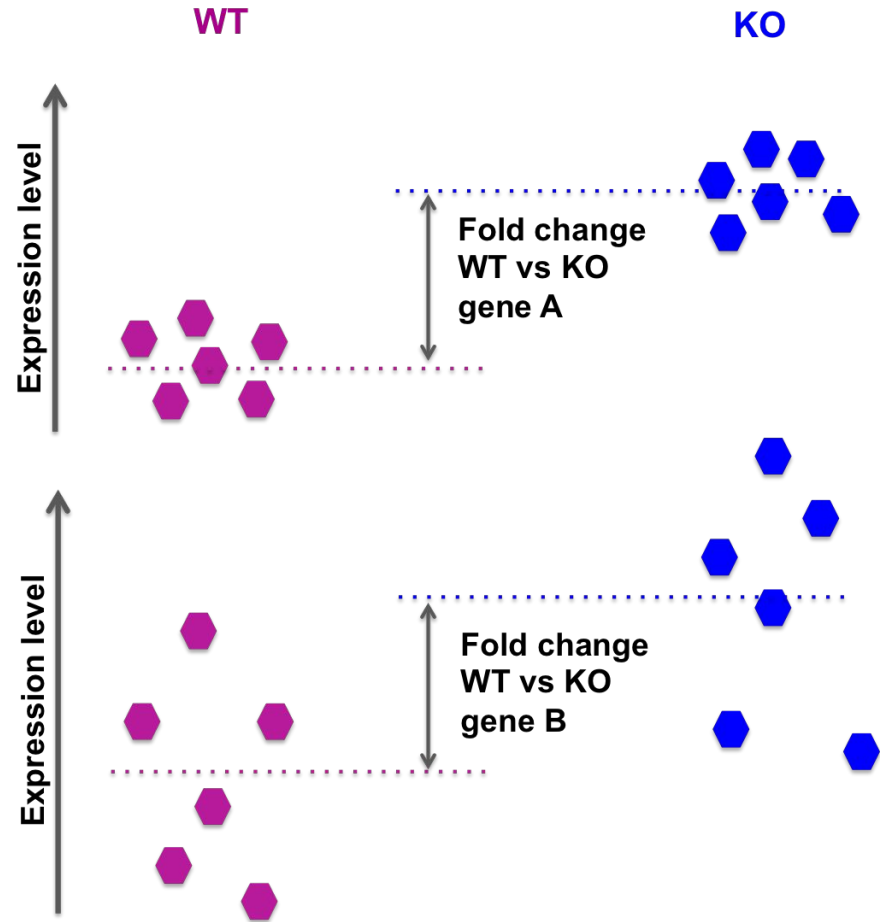
Dataset

What is the fold change?



Dataset

What is the fold change?



Dataset - log2 fold change

- Is the gene more or less expressed in cell senescence compared to control?

Negative log2FC (< 0)
downregulation in
cell senescence

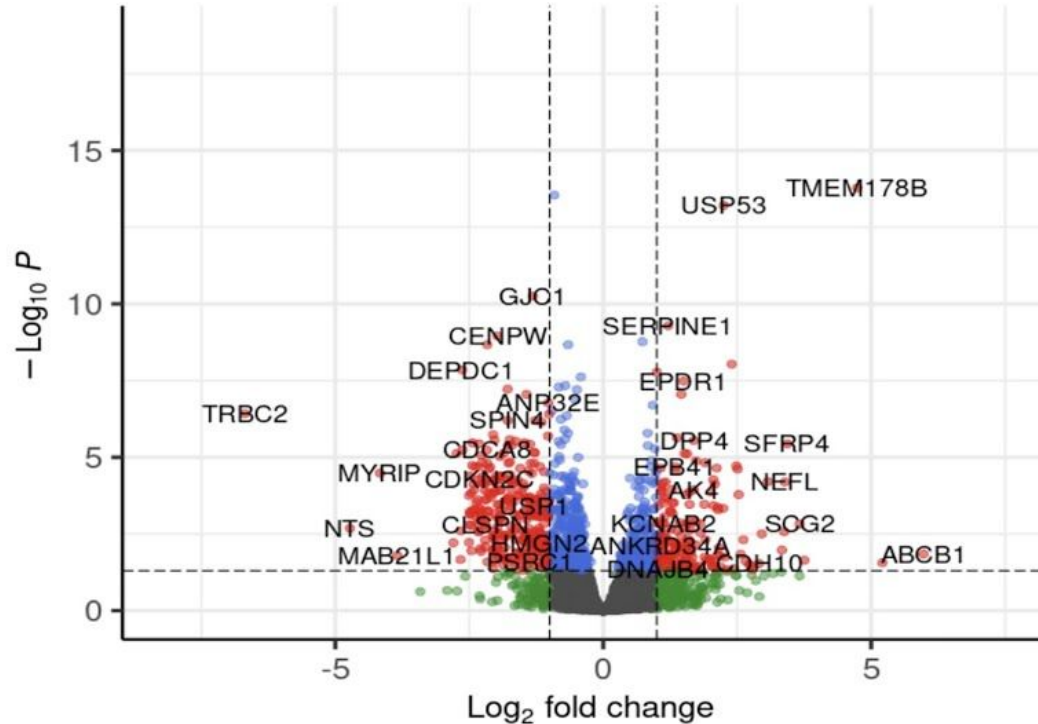
log2FC = 0
both group show same
expression level

Positive log2FC (> 0)
upregulation in
cell senescence

Example: $\log_2\text{FC} = 1 \rightarrow \text{fold change} = 2^1 = 2 = 2\text{-fold increase}$

- gene is twice as expressed in cell senescence compared to control

Dataset - Volcano plot



Typical cutoff values:

$\log_2 \text{FC} = 1.00$

$\log_2 \text{FC} = 0.58$

$\log_2 \text{FC} = 0.26$

Hands-on Module 1 - Explore the dataset

Exploration:

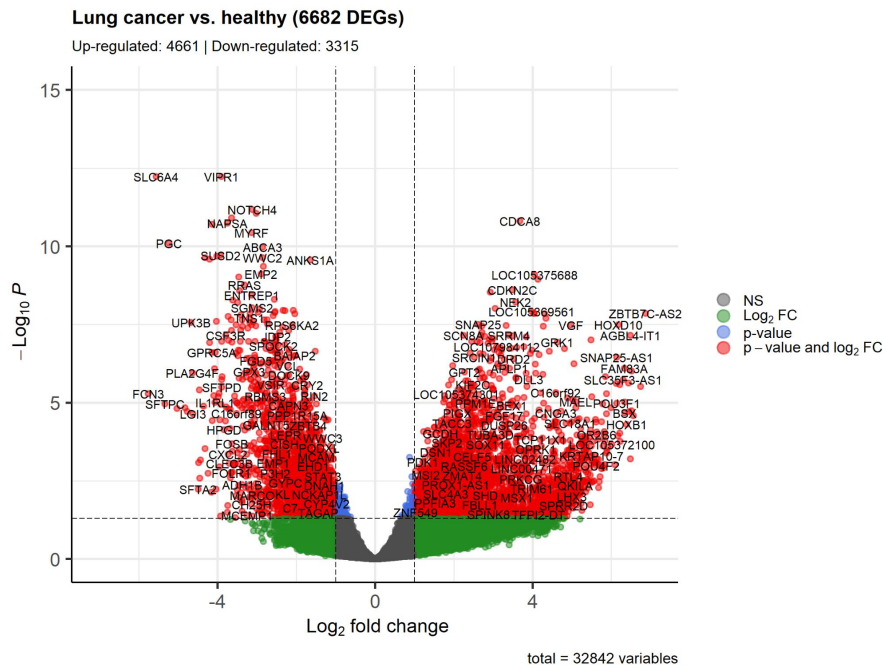
- lung cancer vs. healthy

Questions:

- Number of differentially expressed genes ($|\log_2FC| > 1$ and $p\text{-val} < 0.05$)?
- Number of up-regulated genes?
- Number of down-regulated genes?
- Create the volcano plot → interpretation (discuss with each other)

Key insights

- More up-regulated than down-regulated genes
- We're looking at ~32,000 genes in total and nearly 7,000 genes that are differentially expressed - that's too many to interpret individually
- Volcano plot shows both magnitude (x-axis) and confidence (y-axis) of difference!



Pathway statistics

Enrichment analysis



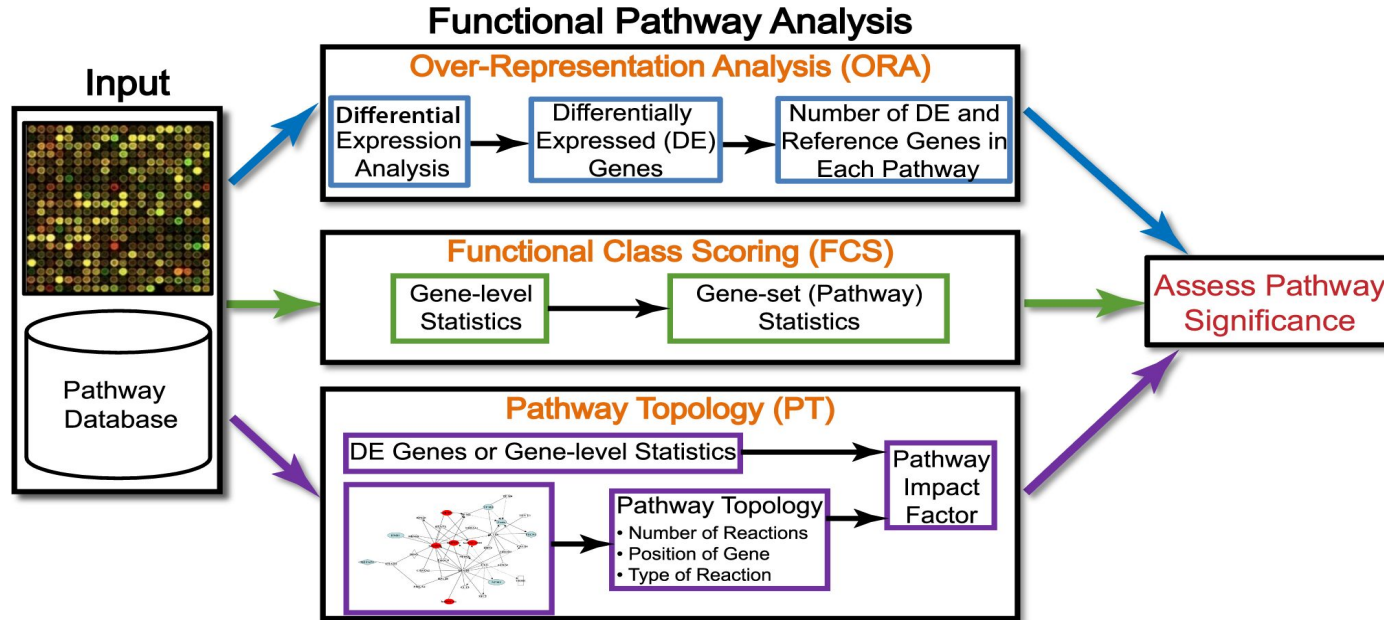
Enrichment analysis

- Aims:
 - Put data into a biological context
 - Intuitive, simple and reproducible analysis
 - Provide a visual representation of the results
 - Reduce complexity from large datasets (thousands of genes) to a set of affected processes

What pathway do we expect to be affected?

- Loss of normal growth control leads to widespread changes
 - Uncontrolled proliferation and survival
 - Resistance to cell death signals
 - Altered metabolism and energy production
 - Tissue invasion and metastatic spread
 - Recruitment of blood vessels (angiogenesis)
 - Evasion of immune surveillance
 - ...

Pathway enrichment analysis



Over-representation analysis

- **Input**

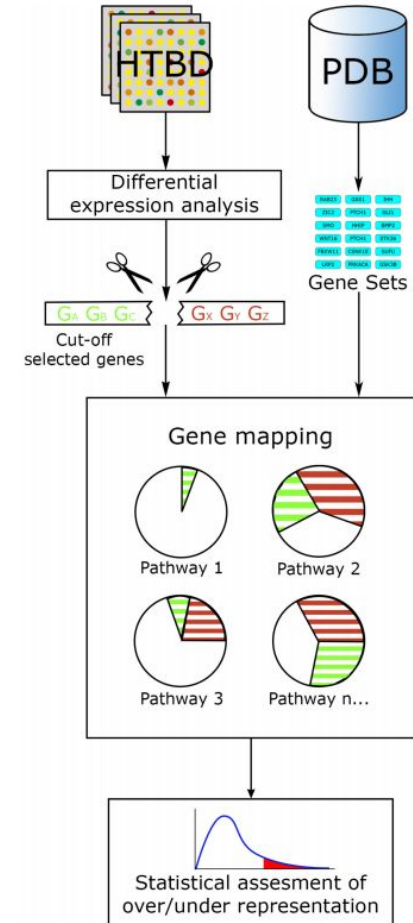
1. Transcriptomics dataset
 - Background = all measured genes
 - Input list = significantly changes genes
2. Pathway database

Over-representation analysis

- **Input**

1. Transcriptomics dataset
 - Background = all measured genes
 - Input list = significantly changes genes
2. Pathway database

- **Statistical test** (hypergeometric test)



Over-representation analysis

- **Input**

1. Transcriptomics dataset
 - Background = all measured genes
 - Input list = significantly changes genes
2. Pathway database

- **Statistical test** (hypergeometric test)

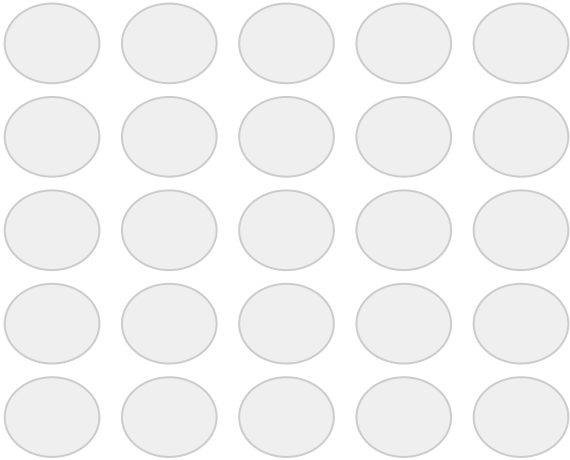
- **Output**

- Enrichment score / p-value per pathway → ranked pathway list

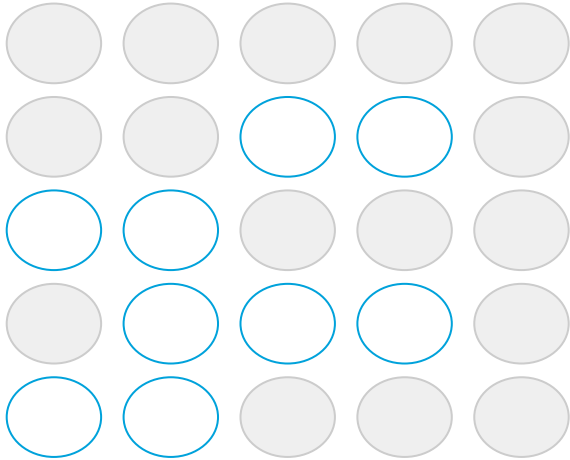
Over-representation analysis

- **N = 25**

background list (total number of measured genes in experiment)



Over-representation analysis



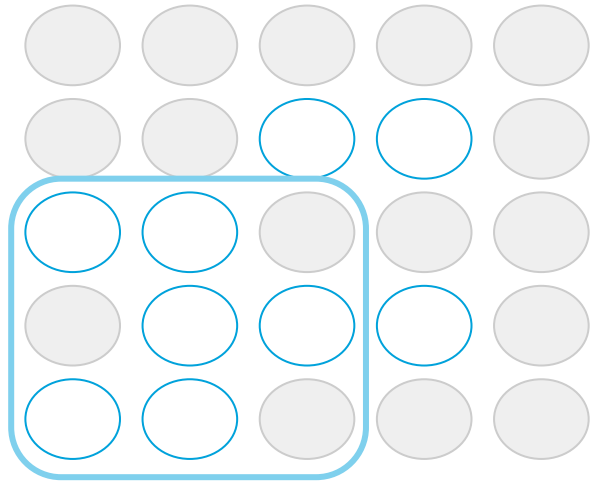
- **$N = 25$**

background list (total number of measured genes in experiment)

- **$R = 9$**

input list (number of changed genes in experiment)

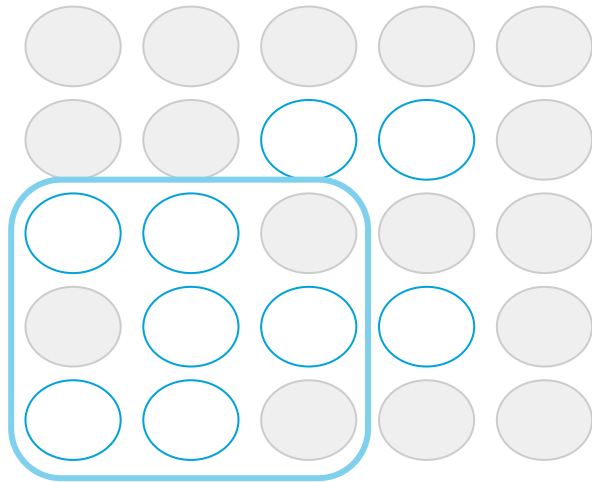
Over-representation analysis



Pathway X

- **$N = 25$**
background list (total number of measured genes in experiment)
- **$R = 9$**
input list (number of changed genes in experiment)
- **$n = 9$**
total number of genes in pathway

Over-representation analysis



Pathway X

- **$N = 25$**
background list (total number of measured genes in experiment)
- **$R = 9$**
input list (number of changed genes in experiment)
- **$n = 9$**
total number of genes in pathway
- **$r = 6$**
number of changed genes in pathway

Pathway analysis results

- Overrepresentation analysis:
 - Ranked list of pathways

Table 1 Seven pathways changed in the diabetic liver

Pathway	Z-score	P-value	# Genes	Genes
Triacylglyceride Synthesis	3.78	0.001	3 / 19	↑ AGPAT2, GPD1, DGAT1
Proteasome Degradation	3.32	0.006	5 / 53	↑ RPN1, PSMB3, HLA-B, HLA-E, HLA-J
Statin Pathway	3.10	0.006	3 / 25	↑ DGAT1, APOA4, CYP7A1
Fluoropyrimidine Activity	2.84	0.013	3 / 28	↑ SLC22A7 ↓ ABCG2, DPYD
Pathogenic Escherichia coli infection	2.76	0.011	4 / 46	↑ ARPC1A, ARPC1B, ACTB ↓ ROCK1
Adipogenesis	2.41	0.016	7 / 121	↑ SREBF1, CDKN1A, NR1H3, PNPLA3, AGPAT2 ↓ CISD1, ZMPSTE24
AMPK Signaling	2.38	0.029	4 / 54	↑ SREBF1, P21 ↓ LEPR, PFKFB3

Pathway statistics in PathVisio revealed seven significantly altered pathways (Z-score > 1.96, P-value < 0.05, minimum of 3 changed genes). The number of genes (# Genes) represent the number of differentially expressed genes in the pathway compared to the total number of measured genes in the pathway. The arrows indicate up ↑ and down-regulation ↓.

Interpretation!

Be aware!!



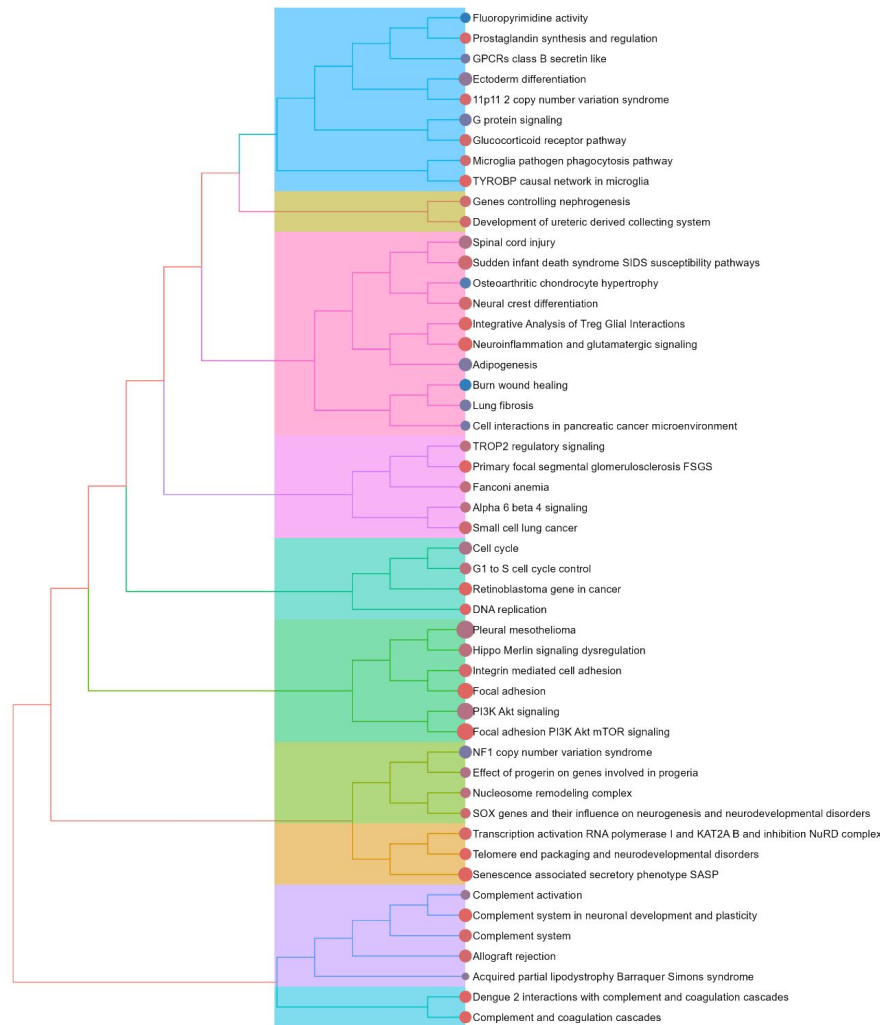
- Be aware!
 - ORA **does not** take pathway topology into account!
 - You don't know yet where the changes occur in the pathway.
 - Always look at the pathway diagrams and study the changes to make the right conclusions!

Hands-on Module 2 - Enrichment Analysis

- Perform pathway statistics to identify which pathways in WikiPathways are significantly affected
- **Questions:**
 - What pathways are affected?
 - Do you see the expected biological pathways dysregulated?

Key insights

- Enrichment asks: do we see more DEGs in this pathway than we'd expect by chance?
- Multiple interconnected pathways characterize SCLC biology
- **Proliferation:** Cell cycle control, G1 to S transition, DNA replication
- **Cell survival & migration:** PI3K-Akt signaling, focal adhesion, integrin-mediated adhesion
- **Neuroendocrine identity:** Neural crest differentiation, neurogenesis
- **Tumor microenvironment:** SASP, complement activation, immune interactions
- **Development:** SOX genes, telomere maintenance, nucleosome remodeling
- ...



Pathway visualization

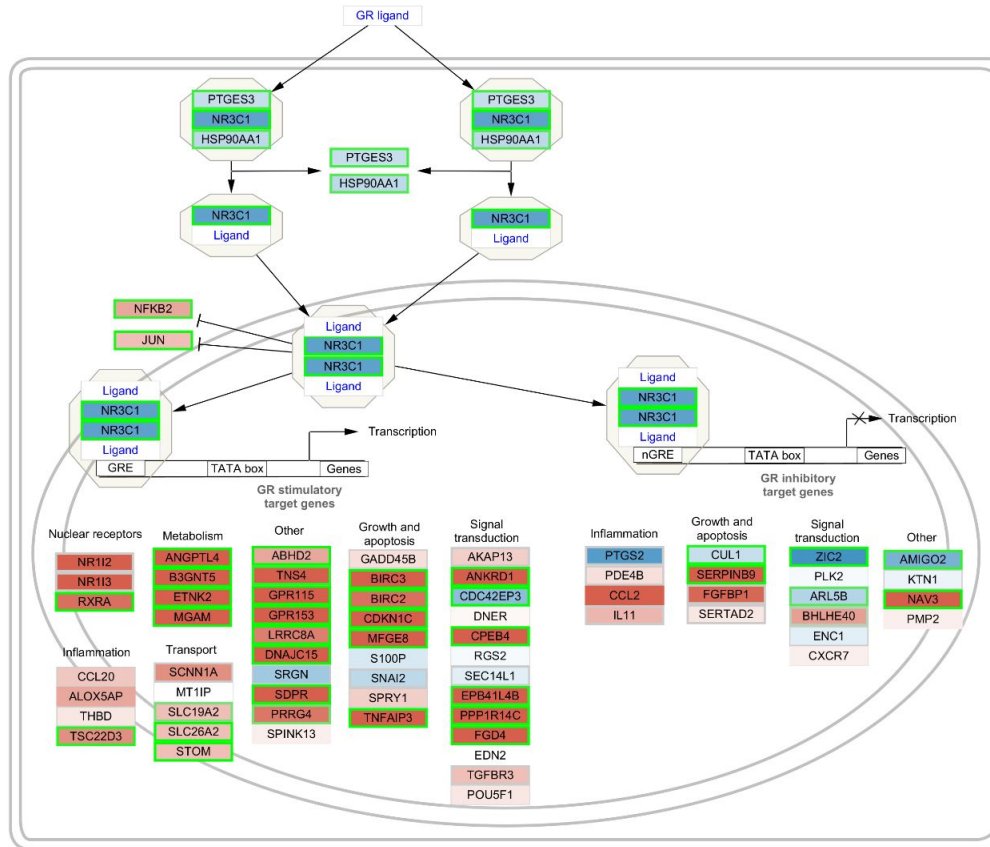


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Cytoscape

1. Open pathway in Cytoscape (example: WP5321)
2. Import data
3. Style pathway

Pathway visualization



Hands-on Module 3 - Pathway visualization

- Make sure you have Cytoscape open
- Script will open pathways in Cytoscape, add data and create a visual style
- **Questions:**
 - Which genes in the pathway are up- or down-regulated?
 - Does this make sense biologically?

- Cell cycle strongly up-regulated
- Complement activation strongly down-regulated



Interactive session

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Summary

- Pathways are **intuitive, machine-readable models** of biological processes
- Pathways can be used to **analyze experimental data** (visualization and statistics)
- **Automation** (e.g. in R) makes an analysis reproducible and can be reused for another dataset
- Pathways are a starting point to build larger maps or executable models

Questions?

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